

Training Report

Comprehensive analysis of model performance and deployment readiness.

Training Analytics Detail

Comprehensive analysis of the training process and model performance

Dataset Details

Information about the data used to train and validate this model.

Item	Value	What it means
Training Epochs	5	The AI studied the data 5 times to learn.
Validation Images	70	Number of images used to test the AI after training.
Total Instances	70	Total number of labeled objects in the validation set.
Image Size	800px	All images were resized to 800×800 pixels.
Classes	1	cat

Box Detection Metrics

How well the AI draws boxes around objects. Higher values = better performance.

PRECISION 90.3% 90 out of 100 detections were correct.	RECALL 98.6% The AI found 99 out of 100 real objects.	F1 SCORE 94.3% Balance between precision and recall.	MAP@50 0.957 Average accuracy of box placement.
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✓ Class-wise Performance

How well the AI performs for each type of object it was trained to detect.

Class	Box Precision	Box Recall	Box F1	Box mAP@50
cat	90.3%	98.6%	94.3%	0.957

🎯 Section 02: Prediction Analytics

TOTAL DETECTIONS

104

Across all images

AVG. CONFIDENCE

65.2%

Mean confidence score

GT COVERAGE

70

Ground truth objects available

PERFORMANCE METRICS (VS GROUND TRUTH)

PRECISION

67.3%

Correctness

RECALL

100.0%

Completeness

F1 SCORE

80.5%

Harmonic Mean

TRUE POSITIVE

70

Correct Detections

FALSE POSITIVES

34

Extra Detections

MISSING (FN)

0

Undetected Objects

Class Distribution		
Class Name	Detections	Distribution
cat	104	100.0%

Confidence Range Breakdown		
Confidence Range	Total Detections	Distribution
0.0-0.2	0	0.0%
0.2-0.5	20	19.2%
0.5-0.8	59	56.7%
0.8-1.0	25	24.0%

Section 03: Evidence-Based Diagnostics

CRITICAL ADJUSTMENTS REQUIRED



The model has significant blind spots that make it unsafe for autonomous operation. Retraining with targeted data is mandatory.

Deep-dive analysis of model failures by object size and spatial location.

3.1 Size Performance (Scale Fidelity)

Size Performance is Optimal

OBSERVATIONS:

- High Consistency: The model performs uniformly across all size groups.
- No major blind spots: Scaling behavior is predictable and stable.



RECOMMENDATIONS:

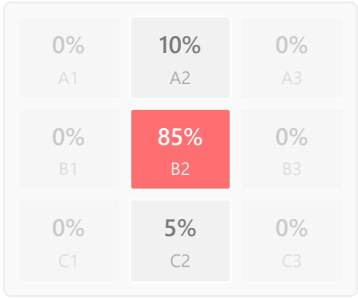
- Maintain quality: The current scale distribution is healthy. No changes needed.

Size Group	Reliability AT 54% TARGET	Samples
Tiny	0%	0
Small	73.7%	19
Medium	100%	26
Large	100%	25

Complex Spatial Blind Spots Identified

OBSERVATIONS:

- ⚠️ Critical Zone B2 (center): 85% of all failures — 3 missed objects and 14 false alarms. Most affected class: "cat".
- Size Breakdown: Tiny: 12 errors (mostly false alarms: 0 missed, 12 false alarms) — danger zone: B2 (center, 91.7%) · Small: 8 errors (mixed errors: 5 missed, 3 false alarms) — danger zone: B2 (center, 75%).
- Class Breakdown: "cat": 20 errors (mostly false alarms: 5 missed, 15 false alarms).



ERROR DENSITY HEATMAP

💡 RECOMMENDATIONS:

- Priority Audit (B2): Manually review "cat" labels in the center area of your images. Focus on over-sensitive detections.

Section 04: Deployment Readiness

Automated stability audit to determine if the model is ready for live production use.

RELIABILITY LIMIT

75%

Maximum Confidence Score

OPTIMAL BALANCE POINT

54%

Recommended Confidence Setting

AUTOMATION RATE

93%

% of real objects found by model

DEPLOYMENT STATUS

READY

Recommended for Pilot

🔍 PRODUCTION ASSESSMENT

- [CEILING] MAXIMUM SAFE CONFIDENCE at ~75%: TP collapses beyond this point.
- [OPTIMAL] 54% confidence (Best F1 Score).
- [RESULTS] 65 correct, 15 false alarms, 5 missed (F1=86.7%).
- [WHY NOT 53%] FP rises faster than TP (more false alarms than value).
- [WHY NOT 55%] FN increases and Automation drops (more missed objects).

⬆ TRAINING ASSESSMENT

- [CEILING] MAXIMUM SAFE CONFIDENCE at ~75% (structural drop zone).
- [TARGET] Pre-wall confidence: 27% (best usable region before collapse).
- [AT 27%] TP=70, FP=31, FN=0 → Automation=100%.
- [IoU] Target IoU: 0.74 (box quality goal for next training).
- [DIAGNOSIS] Improve confidence strength to push ceiling right. Model needs stronger feature extraction.